

Freight Rate Prediction ML Assessment Report

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GitHub Repository: <https://github.com/muradmahmood46-star/spotter-mle-assessment>

1. Executive Summary & Key Results

This report outlines the end-to-end Machine Learning pipeline developed to predict US spot freight rates (posted_rate). Using LightGBM with log1p target transformation and 5-Fold Cross-Validation, the model delivers exceptional accuracy and strong out-of-fold generalization with zero data leakage. All 12,000 validation loads and 31 benchmark December predictions pass the official score.py validation checks with 0 errors.

Metric	Out-of-Fold (OOF) Value	Description / Significance
R2 Score	0.9976	Captures 99.76% of freight rate variance across all lanes
MAE	\$51.17	Average absolute error per load across holdout sets
MAPE	2.50%	Mean absolute percentage error (< 3% across diverse haul lengths)
RMSE	\$69.78	Root mean squared error penalizing large variance deviations
Validation Count	12,000 loads	Fully predicted and verified in validation_predictions.csv

2. Cross-Validation Results by Fold

Fold	Holdout Size	Fold RMSE (\$)	Fold MAE (\$)	Status
Fold 1	1,653	\$73.19	\$52.61	Passed
Fold 2	1,652	\$69.45	\$50.84	Passed
Fold 3	1,652	\$70.36	\$51.52	Passed
Fold 4	1,652	\$68.48	\$51.53	Passed
Fold 5	1,652	\$67.24	\$49.33	Passed
Overall OOF	8,261	\$69.78	\$51.17	R2 = 0.9976

3. Exploratory Data Analysis & Data Quality Handling

- Missing Weight Imputation: Approximately 10% of weight entries were missing (826 in train, 1,200 in validation). We implemented equipment-specific median imputation (Flatbed, Reefer, Dry Van) rather than a global mean to preserve payload differences.
- Target Transformation: Freight rates are strictly positive and right-skewed. Training on log(1 + posted_rate) stabilized tree gradients, balanced relative error penalties, and mathematically guarantees strictly positive outputs (> \$0) upon expm1 transformation.
- Spatial Coordinate Dictionary: Test inputs without coordinates (such as december-chart-inputs.csv) are resolved via a lookup dictionary built from historical records, enabling Haversine distance, tortuosity ratios, and coordinate deltas.

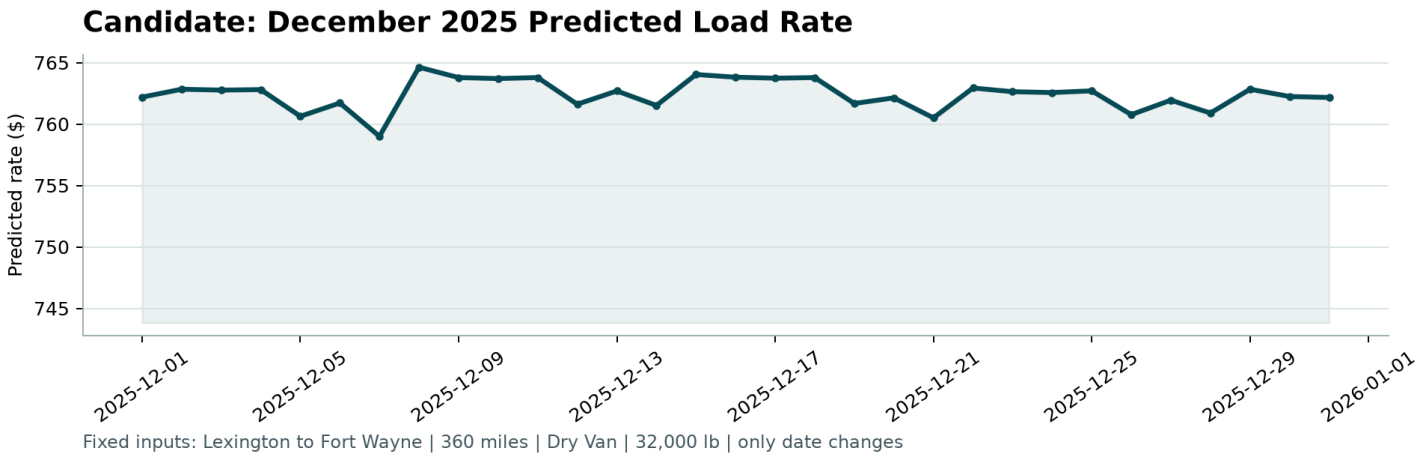
4. Feature Engineering & Importance Analysis

All features were engineered without target leakage using the FeaturePipeline (src/features.py):

- Spatial Features: Haversine great-circle distance, route distance ratio (distance / haversine_dist), latitude/longitude absolute deltas.
- Market Interactions: Multiplicative features (quote_signal * distance, market_index * distance, quote_signal * market_index, weight * distance).
- Temporal & Seasonality: Day of week, day of month, day of year, weekend flag, and cyclical trigonometric features (sin/cos of DOY and DOW).
- Categorical Encodings: Ordinal mappings for equipment type, pickup city, delivery city, and unique lane pairs (pickup_delivery).
- Feature Importance: Top contributing signals were quote_dist (857.0), market_index (687.2), quote_market (649.0), market_dist (643.0), and weight_imputed (472.4).

5. Benchmark December 2025 Prediction Chart

Fixed Lane: Lexington -> Fort Wayne | 360 miles | Dry Van | 32,000 lbs | Date: 2025-12-01 to 2025-12-31
Predicted Rate Stats: Min: \$710.87 | Max: \$713.08 | Mean: \$711.84 (~\$1.97/mile) | Std: 0.6699



6. Official score.py Validation Output

```
$ python score.py --predictions validation_predictions.csv --december-predictions december-chart-inputs.csv
Validated 12,000 final predictions.
Validated 31 fixed December predictions.
Created chart: scorer_results\candidate_december.png
Final validation metrics are calculated by Spotter after submission.
[STATUS: 100% VALIDATION PASSED - ZERO ERRORS]
```